

3.3 Red Blood Cell Parameters

▼ Venepuncture

- ▼ What is the antecubital fossa?
 - The anatomical **space between the forearm and anatomical arm** (around the elbow joint)
 - You can use this region as a landmark for venepuncture
- ▼ What is the name of the piece of equipment wrapped around the arm to stop the flow of blood before venepuncture?
 - **Tourniquet**
 - Use a disposable one
 - The **tourniquet allows the vein to pop up in the arm very nicely**, so we can obtain a blood sample more easily



- ▼ How do we know the needle is actually inside the vein once we have inserted it in a patient for venepuncture?
 - There should be a **flashback in the cannister** (round tube used to contain blood)

- ▼ How do you obtain the small amount of blood from someone to gain an adequate sample for the haematocrit?
 - It is usually safest and most comfortable to **puncture the sides** of the **distal phalanges** (fingertips)
 - Once punctured, **firm pressure** can re-open the wound site
 - For the **haematocrit**, you will use a **capillary tube**
 - For the **MHC and MCHC**, you will use a **Haemocue cuvette**
- ▼ What is the haematocrit?
 - The **percentage by volume of erythrocytes** (red blood cells) **in your blood**.
 - i.e The percentage of the total blood volume that is occupied by erythrocytes
- ▼ How can you measure the haematocrit in a human?
 - Clean surface with alcohol, allow it to evaporate
 - Pierce the skin with a lancet and allow the blood to flow outwards
 - Use capillary tube to collect blood from the incision made by lancet until over 50% full
 - Place capillary tube on wax and allow the end to run dry
 - Centrifuge the blood-containing capillary tube for ten minutes
 - Use the **micro haematocrit reader** to interpret the results of this centrifugation.
- ▼ What is the purpose of centrifuging the blood sample in haematocrit reading?
 - To **separate the red blood cells from the rest of the plasma** in the blood.
 - Naturally, the **red blood cells will accumulate at the bottom of a sample**

▼ Describe how to use the micro haematocrit reader to find the haematocrit of a blood sample.

- Align the bottom of the centrifuge tube with the "zero-line" on the reader
- Slide the adjustable part of the reader to align with the **top of the dark erythrocyte solute component** which has been separated from the plasma
- Read where the line on the reader is pointing to on the y-axis.
- This is your haematocrit.



▼ What unit is used to measure the haematocrit?

- **Litre of red cells per litre of blood cells (L/L)**
- It is usually given as a percentage or decimal (i.e 47% or 0.47)

▼ What is the name of the type of error which is commonly obtained by incorrect alignment of the bottom of the blood sample and the "zero line" on the reader?

- **Parallax error.**

▼ How can you measure the red blood cell count?

- Use a **haemocytometer** to count the number of red blood cells in a particular area
- Multiply this number by the relevant factor to calculate total number of red blood cells in blood

▼ What unit is used to measure red blood cell count?

- **cells/Litre.**

- ▼ What unit is used to measure mean cell volume (mean volume of a blood cell)?
 - **Femtolitres (fL) (10^{-15} L)**
 - MCV is the average volume of each red cell
- ▼ What are two important safety considerations you must make when taking blood from someone?
 - Use alcohol to clean the surface before puncturing (also allow it to evaporate)
 - Wear gloves so that blood does not touch your skin.
- ▼ How can you measure haemoglobin concentration of the blood?
 - Use a cuvette to collect blood from sample
 - Place cuvette in **haemocue device** and read the value that is projected
- ▼ What unit is used to measure haemoglobin concentration?
 - g/L (grams per litre).
 - **Grams of haemoglobin per litre of BLOOD**
- ▼ What is "intention to treat" analysis compared to "per protocol" analysis?
 - **"Intention to treat"** analysis is more tolerant of anomalies than per protocol analysis, as it is used to mirror actual practice and see how clinically effective drugs are, including those who may not take the drug course correctly.
 - It describes how the drug course has the intention to treat the patient's condition but acknowledges that for one reason or another, this may not have happened
 - **"Per protocol"** analysis typically contains "cleaner" data as it removes anomalies and does not account for those who may not have followed instructions when undergoing the drug course. It strictly wants to focus on the data that "matters."
 - It analyses the data of those who have kept per protocol and followed instructions of how to use the drugs
- ▼ How can mean red cell volume (MCV) be calculated?
 - **Haematocrit** (Litres of red cell / Litre of blood) / **red blood cell count** (number of red cells / Litre of blood)
 - This is effectively litres of red cell / number of red cells which will give you a quantity in litres (which we want)

▼ How can mean cell haemoglobin (MCH) be calculated?

- **Haemoglobin concentration (Hb)**(g/L) / **red blood cell count** (red cells/Litre)
- This is effectively haemoglobin amount in grams / number of red blood cells, which calculates the mean amount of haemoglobin per red cell.

▼ How can mean cell haemoglobin concentration (MCHC) be calculated?

- **Haemoglobin concentration (Hb)** (grams of haemoglobin / Litre of blood cells) / **Haematocrit** (Litres of RBC / Litre of blood cells)
- MCHC differs from MCH because it calculates how much haemoglobin